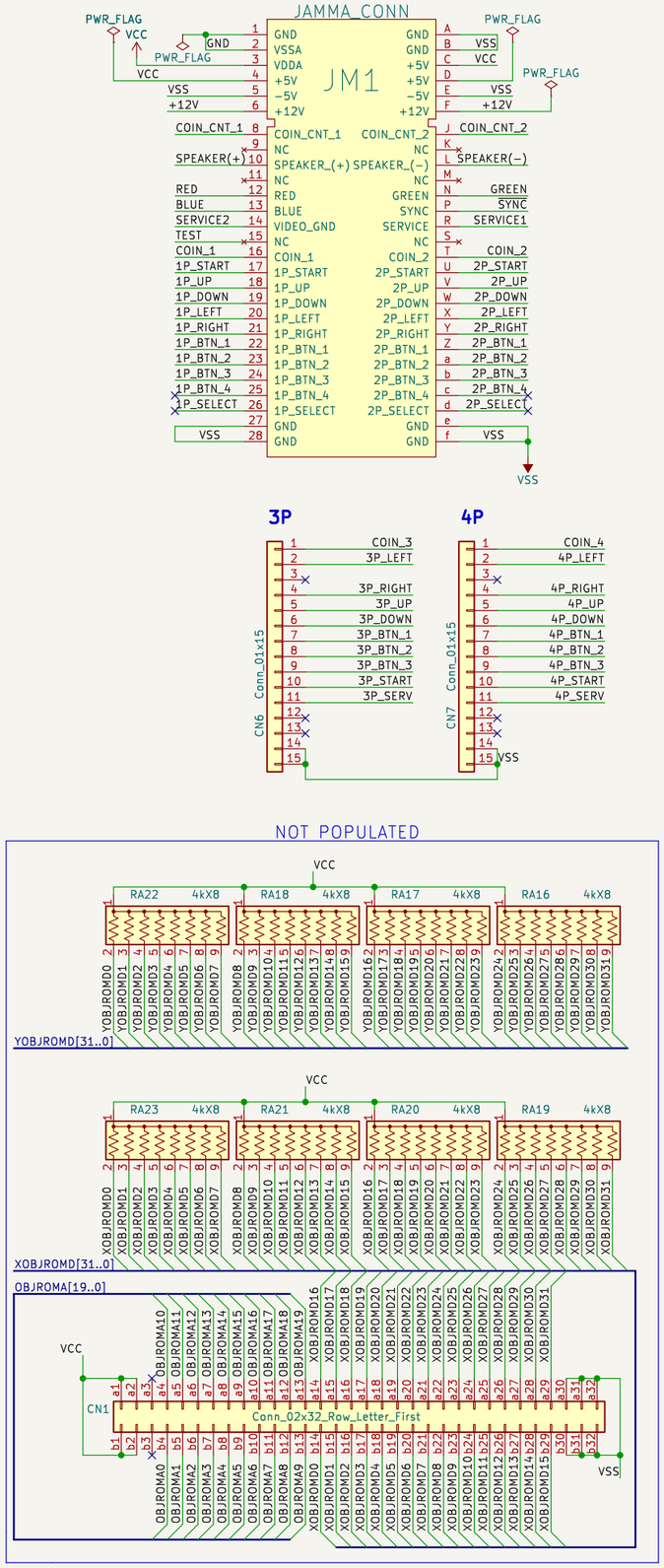
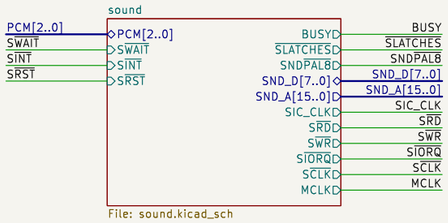
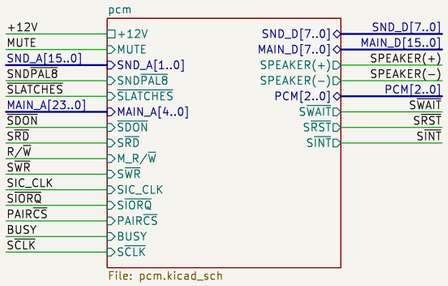
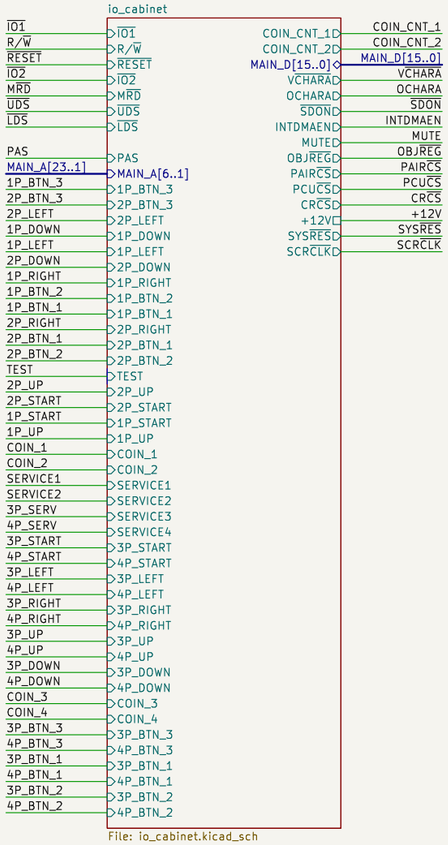
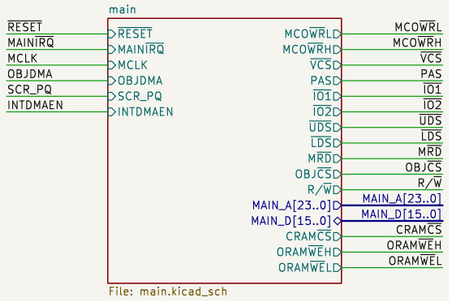
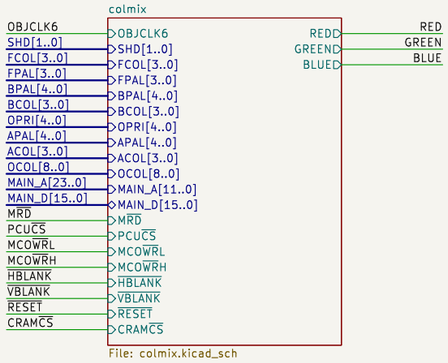




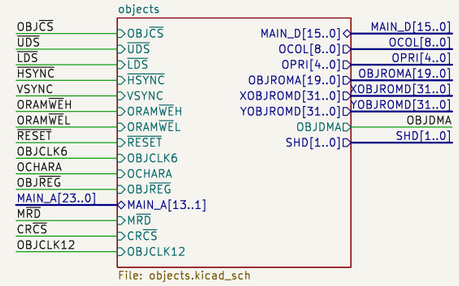


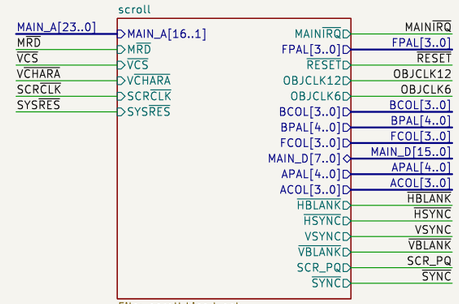










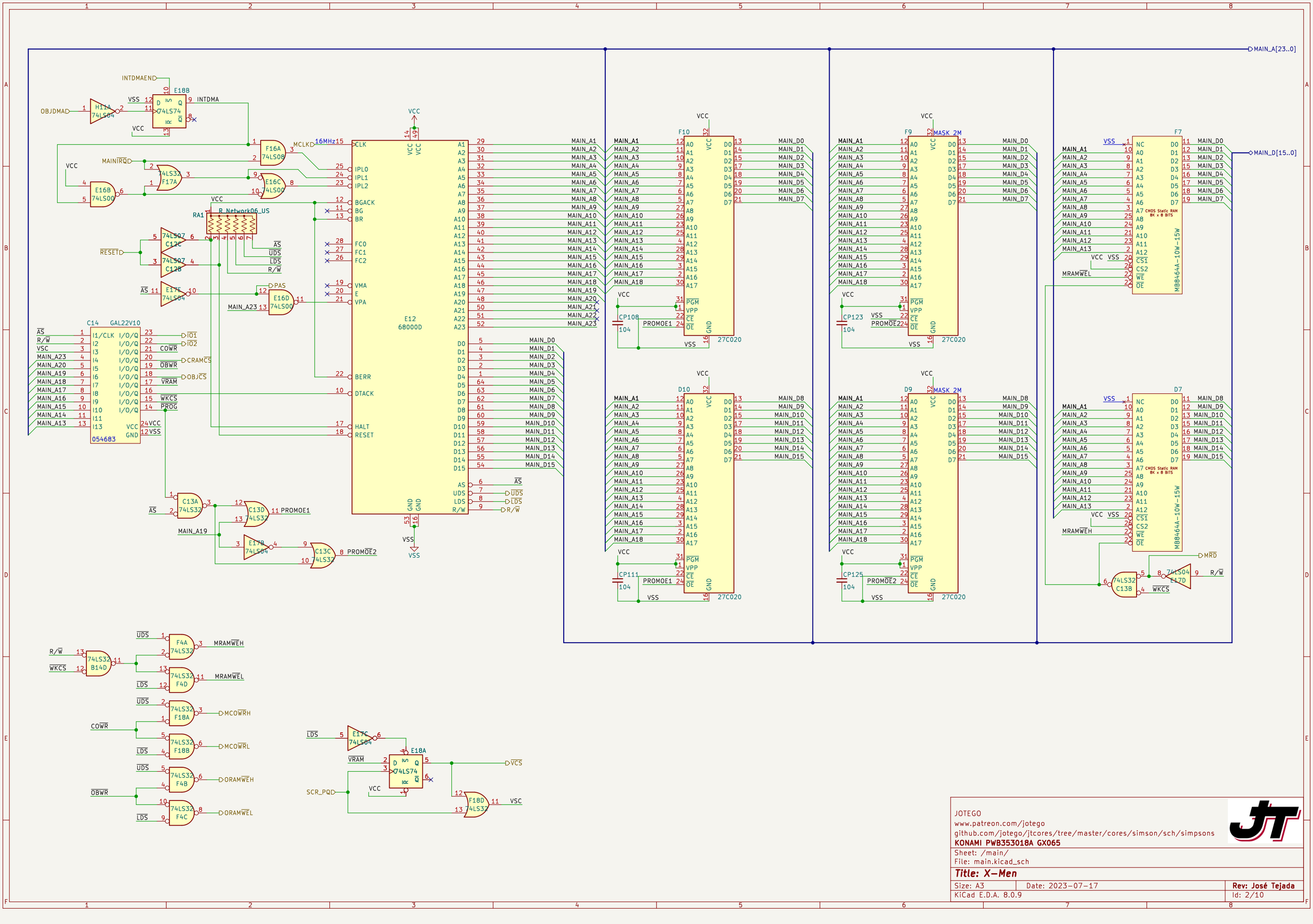


JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /
File: xmen.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 1/10



JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /main/
File: main.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 2/10

S34
3P_BTN_3
4P_BTN_3
COIN_3
COIN_4
R3P_BTN_3
R4P_BTN_3
RCOIN_3
RCOIN_4
KONAMI_005273
S33
3P_BTN_1
4P_BTN_1
3P_BTN_2
4P_BTN_2
R3P_BTN_1
R4P_BTN_1
R3P_BTN_2
R4P_BTN_2
KONAMI_005273
S37
3P_LEFT
4P_LEFT
3P_RIGHT
4P_RIGHT
R3P_LEFT
R4P_LEFT
R3P_RIGHT
R4P_RIGHT
KONAMI_005273
S36
3P_UP
4P_UP
3P_DOWN
4P_DOWN
R3P_UP
R4P_UP
R3P_DOWN
R4P_DOWN
KONAMI_005273
S40
1P_BTN_3
2P_BTN_3
COIN_1
COIN_2
R1P_BTN_3
R2P_BTN_3
RCOIN_1
RCOIN_2
KONAMI_005273
S39
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
R1P_BTN_1
R2P_BTN_1
R1P_BTN_2
R2P_BTN_2
KONAMI_005273
S43
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
R1P_LEFT
R2P_LEFT
R1P_RIGHT
R2P_RIGHT
KONAMI_005273
S42
1P_UP
2P_UP
1P_DOWN
2P_DOWN
R1P_UP
R2P_UP
R1P_DOWN
R2P_DOWN
KONAMI_005273
C17
R3P_BTN_1
R4P_BTN_1
R3P_BTN_2
R4P_BTN_2
RCOIN_3
RCOIN_4
R3P_BTN_3
R4P_BTN_3
MAIN_A1
JOYSTICK
MAIN_D12
MAIN_D13
MAIN_D15
MAIN_D14
74LS257
C18
R3P_UP
R4P_UP
R3P_DOWN
R4P_DOWN
R3P_RIGHT
R4P_RIGHT
R3P_LEFT
R4P_LEFT
MAIN_D8
MAIN_D9
MAIN_D11
MAIN_D10
74LS257
C15
R1P_BTN_1
R2P_BTN_1
R1P_BTN_2
R2P_BTN_2
RCOIN_1
RCOIN_2
R1P_BTN_3
R2P_BTN_3
MAIN_D4
MAIN_D5
MAIN_D7
MAIN_D6
74LS257
CC15
R1P_UP
R2P_UP
R1P_DOWN
R2P_DOWN
R1P_RIGHT
R2P_RIGHT
R1P_LEFT
R2P_LEFT
MAIN_D0
MAIN_D1
MAIN_D3
MAIN_D2
74LS257
MAIN_D[15..0]
TEST
SW17
EVQ-QS5-07K
switch
button
R13
220Ω
R14
3.3KΩ
VCC
CAB14
CC3
104
VSS
CAB[15..0]
S44
1P_START
2P_START
3P_START
4P_START
CAB8
CAB9
CAB10
CAB11
KONAMI_005273
S54
SERVICE1
SERVICE2
SERVICE3
SERVICE4
CAB0
CAB1
CAB2
CAB3
KONAMI_005273
C16
CAB8
CAB9
CAB10
CAB11
CAB14
OTHER
MAIN_D8
MAIN_D9
MAIN_D10
MAIN_D11
MAIN_D14
MAIN_D15
3-StateHexBuffers
74LS367
VCC
R33
4.7KΩ
SW-B12
OFF
SLIDESW1P
NOT
MOUNTED
VSS
D16
CAB0
CAB1
CAB2
CAB3
CAB6
CAB7
OTHER
MAIN_D0
MAIN_D1
MAIN_D2
MAIN_D3
MAIN_D6
MAIN_D7
3-StateHexBuffers
74LS367
D17
MAIN_D8
MAIN_D9
MAIN_D10
MAIN_D11
OCHARA
VCHARA
SDON
MUTE
74LS175
UDS
RESET
74LS32
F17D
CC13
MAIN_D0
MAIN_D1
MAIN_D2
MAIN_D3
MAIN_D4
MAIN_D5
INTDMAEN
74LS174
hex flip flop
13C label is
duplicated on
the PCB
B13
ER5911
EEPROM
CAB6
CAB7
VSS
FL-B12
VCC
+12V
VSS
A11
AFE
SJO
CP76
C_Small
VSS
RESET
CNT2
CNT2
CNT1
CNT1
COIN_CNT_2
COIN_CNT_1
N.C
VCC
VREF
RES
CT
VS
OUT1
OUT2
+5V
+12V
Master/RESET generation
for the whole system
and watchdog timer
KONAMI
051550
G18A
74LS74
SYSRES
G12
24MHz
XTAL_OSC
VCC
VSS
H11C
74LS04
SCRCLK
MAIN_A1
E17A
74LS04
MAIN_A6
MAIN_A5
IO1
A1 D15B O0
A0 O1
O2
E 74LS139 O3
decoder
PAIRCS
PCUCS
MAIN_A[6..1]
MAIN_A3
MAIN_A2
A1 D15A O0
A0 O1
O2
E 74LS139 O3
decoder
JOYSTICK
OTHER
CRCS
MRD
IO2
R/W
B14C
74LS32
B14A
74LS32
74LS32
B14B
LDS
OBJREG
VSS
PAS
F5B
74LS74
AFE
VCC
JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065
Sheet: /io_cabinet/
File: io_cabinet.kicad_sch
Title: X-Men
Size: A3
Date: 2023-07-17
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 3/10

NOT POPULATED

VCC

RA24 4.7kX8 — GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7

RA25 4.7kX8 — GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15

RA26 4.7kX8 — GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23

RA27 4.7kX8 — GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29 GFXROMD30 GFXROMD31

GFXROMA[18..0]

DIN64SS
CNS
Connect Most Letter First

GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7 GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15 GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23 GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29 GFXROMD30 GFXROMD31

J17

GFXROMA0 GFXROMA1 GFXROMA2 — D0 D1 D2 D3 D4 D5 D6 D7 — O0 O1 O2 O3 O4 O5 O6 O7 — GFXROMA'0 GFXROMA'1 GFXROMA'2

VSS VDE VSS — LE OE — LATCHES AND FLIP FLOPS

74LS373

H17

GFXROMA11 GFXROMA12 GFXROMA13 GFXROMA14 GFXROMA15 GFXROMA16 GFXROMA17 GFXROMA18 — D0 D1 D2 D3 D4 D5 D6 D7 — O0 O1 O2 O3 O4 O5 O6 O7 — GFXROMA'11 GFXROMA'12 GFXROMA'13 GFXROMA'14 GFXROMA'15 GFXROMA'16 GFXROMA'17 GFXROMA'18

VSS — LE OE — LATCHES AND FLIP FLOPS

74LS373

OBJCLK12 — H11B 74LS04 — VDE — G18B D IS Q 74LS74 IR Q — VCC

GFXROMA[18..0]

L15 065A08 KONAMI MASK BM - 16BIT — CMOS ROM — 054903

A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16 A17 A18 A19 OE? BYTE? VSS — D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 D14 D15/A-1

GFXROMA'0 GFXROMA'1 GFXROMA'2 GFXROMA'3 GFXROMA'4 GFXROMA'5 GFXROMA'6 GFXROMA'7 GFXROMA'8 GFXROMA'9 GFXROMA'10 GFXROMA'11 GFXROMA'12 GFXROMA'13 GFXROMA'14 GFXROMA'15 GFXROMA'16 GFXROMA'17 GFXROMA'18

L16 065A08 KONAMI MASK BM - 16BIT — CMOS ROM — 054903

GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7 GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15 GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23 GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29 GFXROMD30 GFXROMD31

GFXROMD[31..0]

APAL[4..0] — ACOL[3..0] — SCRA

APAL1 APAL0 APAL4 APAL3 APAL2 ACOL3 ACOL2 ACOL1 ACOL0 VBLANK HBLANK HSYNC CSYNC FCOL0 FCOL1 FCOL2 FCOL3 FPAL0 FPAL1 FPAL2 FPAL3

FIX — FCOL[3..0] — FPAL[3..0]

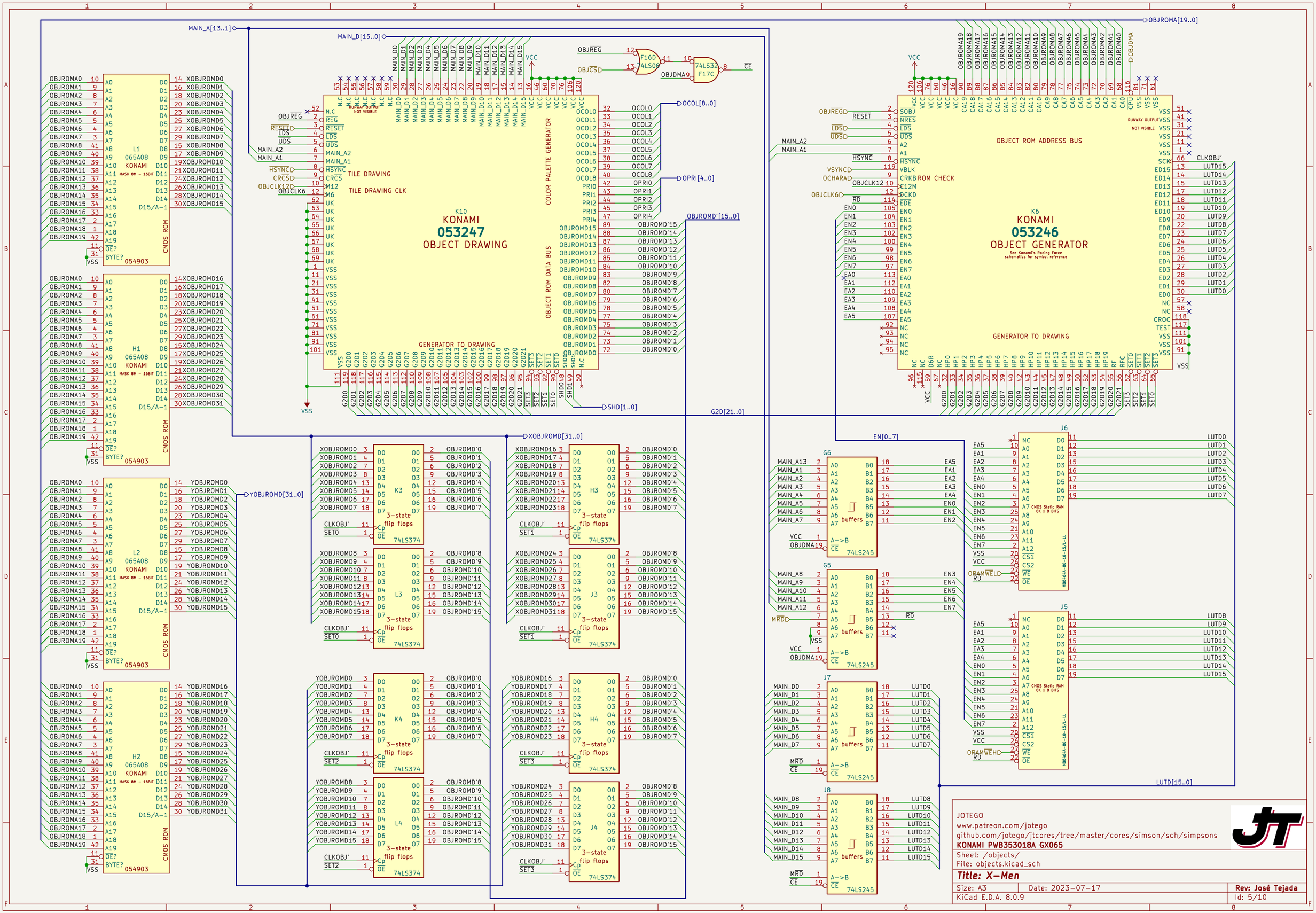
H11D 74LS04 — VSYNC

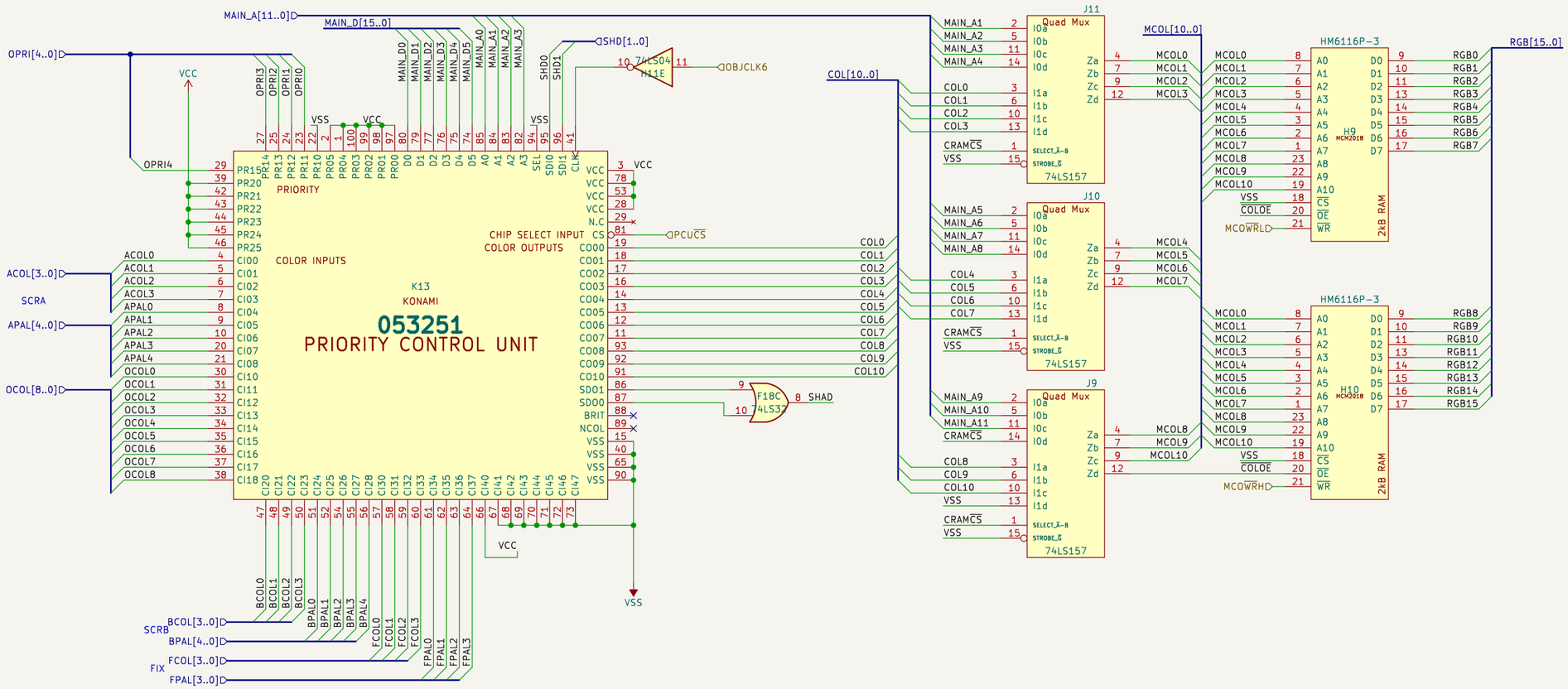
VCC

J14
KONAMI
051962
TILE DRAWING

GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7 GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15 GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23 GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29

VC0 VC1 VC2 VC3 VC4 VC5 VC6 VC7 VC8 VC9 VC10 VC11 VC12 VC13 VC14 VC15 VC16 VC17 VC18 VC19 VC20 VC21 VC22 VC23 VC24 VC25 VC26 VC27 VC28 VC29 VC30 VC31 TEST RESET AB0 AB1 M24 ZA4H ZA3H ZA2H ZA1H ZB4H ZB3H ZB2H ZB1H RMRD CRCS BE RD WR COL7 COL6 COL5 COL4 COL3 COL2 COL1 COL0 VSS VSS VSS


MAIN_A[13..1]
MAIN_D[15..0]
OBJROMA[19..0]
OCOL[8..0]
OPRI[4..0]
OBJROMD'[15..0]
SHD[1..0]
G2D[21..0]
XOBJROMD[31..0]
YOBJROMD[31..0]
EN[0..7]
LUTD[15..0]
K10
KONAMI
053247
OBJECT DRAWING
TILE DRAWING
TILE DRAWING CLK
COLOR PALETTE GENERATOR
OBJECT ROM DATA BUS
GENERATOR TO DRAWING
K6
KONAMI
053246
OBJECT GENERATOR
See Konami's Racing Force schematics for symbol reference
OBJECT ROM ADDRESS BUS
ROM CHECK
GENERATOR TO DRAWING
L1
L2
H1
H2
K3
K4
L3
L4
H3
J3
H4
J4
G5
G6
J5
J6
J7
J8
065A08
KONAMI
MASK 8M - 16BIT
054903
CMOS ROM
74LS374
74LS245
3-state flip flops
buffers
JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065
Sheet: /objects/
File: objects.kicad_sch
Title: X-Men
Size: A3
Date: 2023-07-17
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 5/10



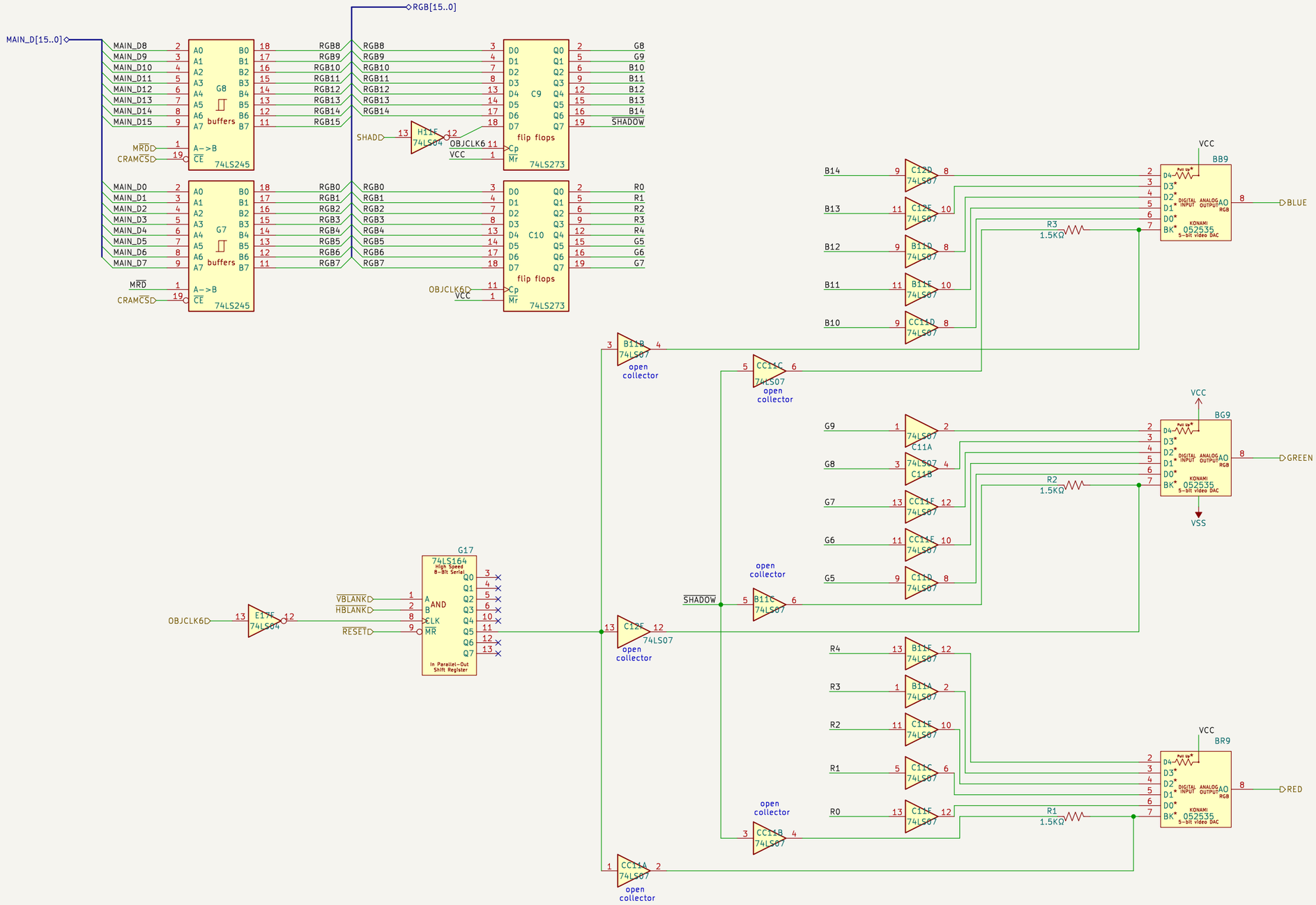
JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons

KONAMI PWB353018A GX065

Sheet: /colmix/
File: colmix.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 6/10

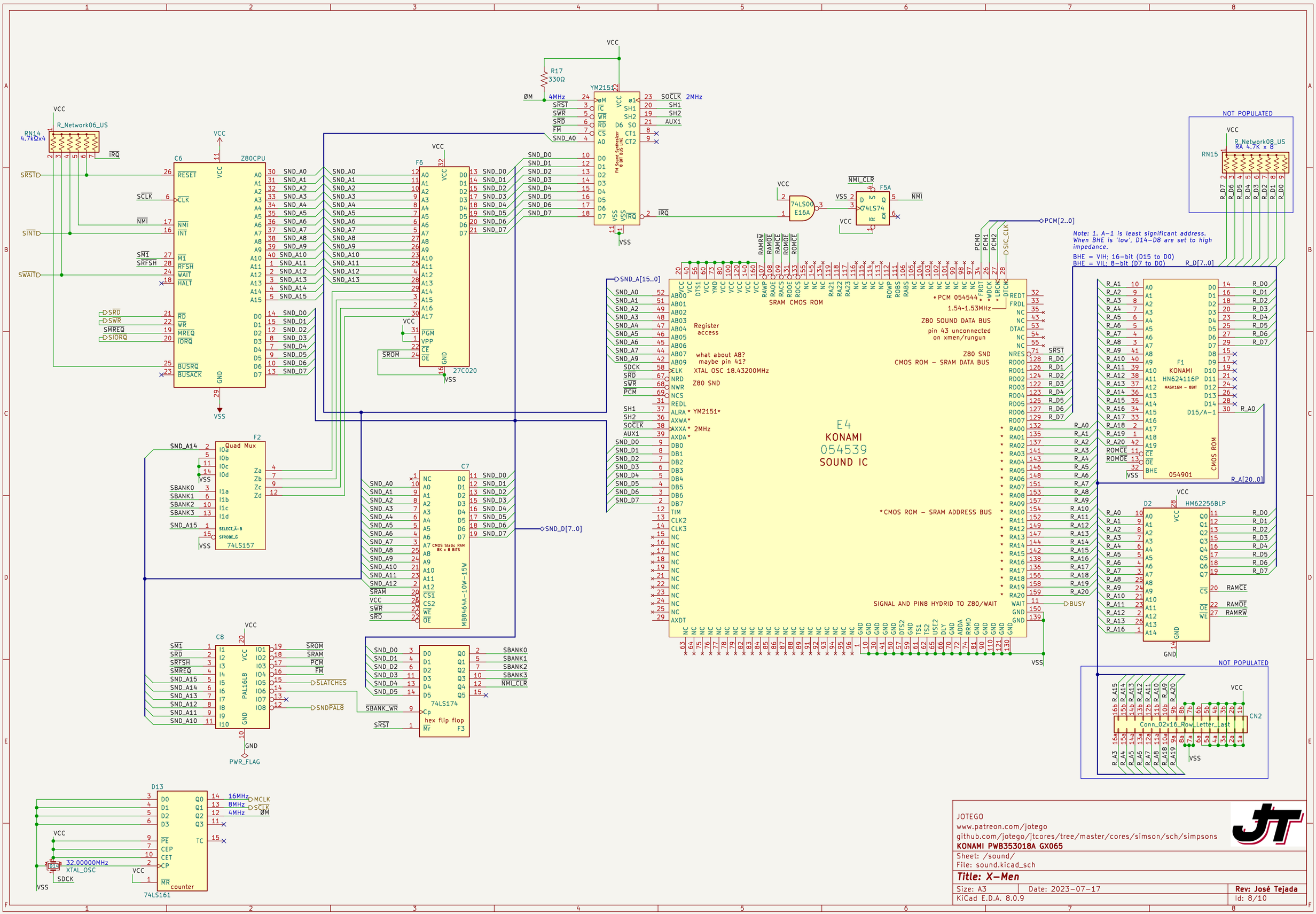


JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /colmix/rgb/
File: rgb.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 7/10



JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons

KONAMI PWB353018A GX065

Sheet: /sound/
File: sound.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 8/10

KONAMI 054544

Audio module

Sound communication latch/volume manager chip

THIS CONNECTOR REPLACES THE PINS OF THE KONAMI CI 054544

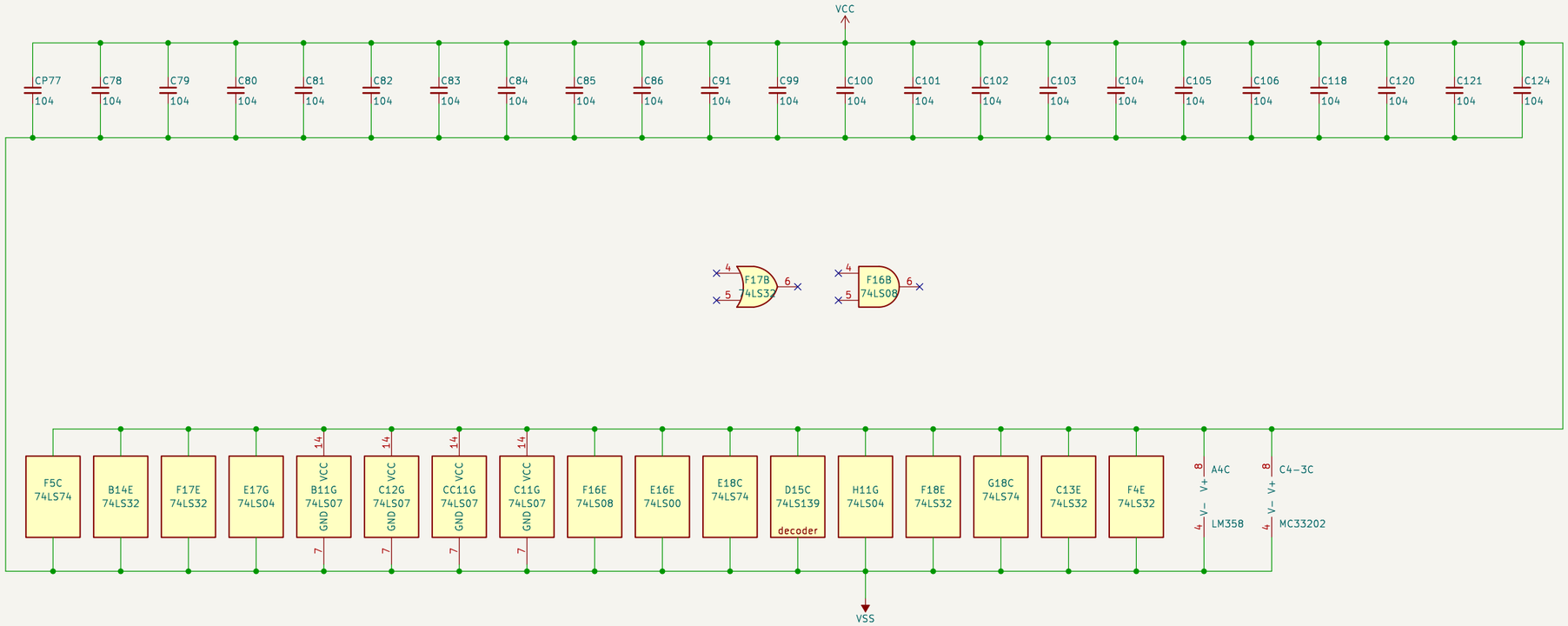
SHORTED ON MOTHER PCB

Pins 8 and 16 of the AD1868R are connected to pins 40 and 36 of the 054544 respectively, but these pins (40 and 36) are floating.

SOUND AMPLIFIER MOTHER PCB

JOTEGO	
www.patreon.com/jotego	
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons	
KONAMI PWB353018A GX065	
Sheet: /pcm/	
File: pcm.kicad_sch	
Title: X-Men	
Size: A3	Date: 2023-07-17
KiCad E.D.A. 8.0.9	Id: 8/10

Rev: José Tejada



JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB353018A GX065

Sheet: /capacitors/
File: capacitors.kicad_sch

Title: X-Men

Size: A3	Date: 2023-07-17	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 9/10